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Published in:
PerDis 2026 - Proceedings of the 10th ACM International Symposium on Pervasive Displays

DOI (link to publication from Publisher):
10.1145/3797993.3798017

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Publication date:
2026

Document Version
Publisher's PDF, also known as Version of record

Link to publication from Aalborg University

Citation for published version (APA):
Nguyen, D. V., Davidsen, F. B., Gad, J. T., Hørlück, K., Aslan, I., & Merritt, T. (2026). The Museum as a Playground: Escape Room in a Submarine Setting. In F. Alt, K. Vaananen, A. Colley, M. Pakanen, A. Wiethoff, & M. Hoggenmuller (Eds.), *PerDis 2026 - Proceedings of the 10th ACM International Symposium on Pervasive Displays* (pp. 1-10). Article 5 Association for Computing Machinery (ACM). https://doi.org/10.1145/3797993.3798017

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The Museum as a Playground: Escape Room in a Submarine Setting

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Abstract

We report on the design of an interactive physical experience aimed at engaging young adults at a maritime museum through an escape room situated inside a real submarine. The intent is to attract more young adults to experience the museum's artifacts. By leveraging the setting of the submarine as a historical maritime artifact and integrating pervasive interaction technologies enabling conversational and tangible interactions, the escape room seeks to transform the traditional storytelling into an immersive and playful educational experience. The design process was based upon cultural participation data, user-centered design methods, and iterative testing, culminating in the creation of a modular, portable setup using a historically accurate submarine suitcase as the central game component. Through playtesting and case analysis, the project demonstrates the potential for escape room experiences to attract younger audiences by fostering exploration, collaboration, and engagement.

Keywords

HCI, museum engagement, interaction design, tangible interaction, escape room design, submarine experience, conversational user interfaces, immersive storytelling, young adult audiences, playful and narrative design

ACM Reference Format:
Diem Vi Nguyen, Frida Brockie Davidsen, Johan Tonsberg Gad, Kristine Hørlück, Ilhan Aslan, and Timothy Merritt. 2026. The Museum as a Playground: Escape Room in a Submarine Setting. In *International Conference on Pervasive Displays (PerDis '26), March 16–18, 2026, Munich, Germany*. ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3797993.3798017

1 Introduction

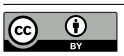
With innovations in pervasive interaction technologies, there is a growing opportunity for museums to explore how they can augment their unique assets with pervasive technology to meet younger generations' expectations and keep delivering socially engaging cultural experiences. Museums are known for providing contextual learning experiences [10]. They are often uniquely equipped to offer intriguing artifacts and spaces to explore a profound interactive experience, including the purposes of enjoyment besides learning [6]. To achieve this, common approaches include gaming and storytelling [28], and the adoption of serious gaming [27].

In this paper, we report on a collaboration with our local maritime museum exploring the special case of a submarine as a historical physical artifact and interaction space. The submarine is one of the museum's main attractions, aiming to remind and inform visitors and young adults of our history and culture. Among several concepts discussed with the museum as a stakeholder to engage especially younger adults, one idea stood out, an escape room inside the submarine. Escape rooms are essentially games in which participants have to complete challenges and solve puzzles [32] and are also used in classrooms and education [29–31].

The following question has guided our research and design activities: *RQ:* How might an interactive escape room experience be designed inside a submarine, to attract young adults by combining immersive storytelling, social play, and maritime history?

Implementing an escape room game builds on the museum's existing assets while also tapping into the social, immersive, and mentally stimulating experiences that define escape rooms [2].

We use an *embodied interaction* lens to ground how "context" matters in this work. Following Dourish, interaction is not separable from the situated, bodily, and material circumstances in which it unfolds [15]. In museum settings, this situatedness is also *place-based*: the physical environment is not merely a container for activity, but a historically and socially meaningful place that shapes what actions feel possible and appropriate [11, 17]. In our case, the submarine is a highly constrained interior with strong material cues (tight passages, low ceilings, equipment layout) and an embedded cultural narrative (Cold War maritime readiness). We therefore define the **submarine context** as the combination of (i) *material constraints* (e.g., limited movement and visibility), (ii) *heritage meaning* (the submarine as an authentic artifact), and (iii) *social configuration* (small-group coordination under time pressure). This lens enables us to interpret participant reports such as "feeling like active crew members" not as incidental comments, but as outcomes shaped by embodied navigation, shared attention, and narrative prompts embedded in the space.



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PerDis '26, Munich, Germany
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ACM ISBN 979-8-4007-2513-5/26/03
https://doi.org/10.1145/3797993.3798017

The project was developed with a local maritime museum whose experience center includes a decommissioned submarine that visitors can enter and explore[1]. The museum attracts approximately 30,000 visitors annually, with attendance peaking during the summer tourist season. However, the museum reported a gap in engagement among young adults compared to children and family groups, motivating an intervention that could (a) leverage the submarine as an authentic setting, while (b) meeting operational constraints such as rapid setup/reset and minimal modification of heritage assets.

This article makes the following contributions:

- Design specifications are provided, which were created in cooperation with a museum for turning a submarine into an escape room with pervasive interaction technologies for the purpose of new forms of storytelling and attracting younger adults.
- Details of an escape room implementation are provided, focusing on the historical role of submarine suitcases.
- We report on our iterative design process and evaluation results, including observations of 12 participants playing the escape room game in the submarine.

2 Related Work

2.1 Playful Participation and Interactive Museum Ecologies

HCI research has long argued that museum experiences emerge through situated participation rather than one-way information delivery. Interactive exhibits can support meaning-making by inviting exploration, conversation, and shared attention around artifacts [10, 11]. Recent reviews consolidate this direction, showing that play can contribute to engagement and learning outcomes while foregrounding practical constraints in public settings (e.g., accessibility, visitor flow, robustness, and maintenance) [26]. Tangible and embodied interaction has been shown to scaffold collaborative engagement in front of exhibits, for example, through tangible tokens and hands-on manipulation that externalizes interpretation and coordination within the group [23]. Museum visits are often social, and combining personal devices with situated displays can structure shared interaction and coordination [7]. We build on this body of work by treating the submarine as the primary interactive asset: an authentic place whose material constraints and atmosphere can be activated through structured, collaborative play.

2.2 Escape Rooms and Pervasive Games in Cultural Heritage and Learning

Escape rooms have expanded beyond entertainment into educational and cultural contexts because they combine narrative framing with time-bounded, collaborative problem solving [29–32]. Within museums, escape-room-inspired exhibitions show how puzzles can scaffold attention, curiosity, and group coordination around exhibits [5]. Pervasive cultural heritage games demonstrate how narrative missions and location-situated challenges can promote engagement across a site, while raising design concerns around orchestration, group progression, and the integration of content into place [13]. Escape-room logic has also been used in public and civic contexts as a structured format for sensemaking and collaboration across physical environments, underscoring the genre's flexibility as a pervasive interaction form [19]. However, designing in authentic heritage environments introduces additional constraints of preservation, safety, staff workload, and reset time.

2.3 Puzzle Infrastructures for Tangible, Resettable, and Place-based Play

A recurring design challenge is achieving puzzle variety and pacing without undermining narrative coherence. Escape-room puzzle taxonomies provide a vocabulary for balancing analog and digital mechanics and reasoning about player progression [21]. In heritage contexts, situated tangible gamification has been shown to support collaborative learning by distributing interaction across bodies, artifacts, and environment [25]. For public installations, research further emphasizes operational considerations: authoring and maintaining interactive installations often requires tooling and practices that support rapid content updates and reliable operation across deployments [22]. Frameworks for situated/projection-based installations similarly foreground robustness and contextual fit [16]. These perspectives motivate our use of a portable physical "hub" (the suitcase) that concentrates electronics into a self-contained unit while orchestrating discovery across the submarine. In contrast to approaches that primarily engage visitors through digital overlays and augmentation [4, 12], we emphasize embodied, place-based exploration and group coordination structured through an escape-room format.

3 Requirements and Design Rationale

To explore both the practical and theoretical aspects of designing an escape room that caters to young adults, providing a prototype for reimagining cultural engagement in a modern museum setting, we identified through multiple meetings, visits, and in coordination with the museum design challenges and requirements, from which we inferred design rationales.

In conversations with the museum director, we identified five core requirements: to support learning through play by building on the popularity of the mini marina with remote-controlled ships; to increase engagement among young adults through more interactive, distinctive experiences; to replace scattered, outdated screens with a more cohesive and updated visualization of the city's development; to reduce reliance on text-heavy, primarily local-language signage by enabling easier translations and more story-driven presentation; and to expand the museum's interactive offer beyond existing installations (e.g., the wind simulator) to better convey the embodied experience of sailing.

The escape room concept is driven by museum constraints and audience goals. It must be quick to install and remove (within 30 minutes) to minimize operational disruption. The design leverages the submarine as an existing narrative-rich artifact, balancing immersive interaction with preservation and participant safety in a less controlled environment than purpose-built escape rooms [14]. To appeal to young adults, it adopts a familiar, social escape room format and follows common conventions (2–6 participants, 60 minutes) to ensure recognisability and pacing [3]. Finally, an authentic submarine suitcase anchors the experience while constraining the

[1] https://springeren-maritimt.dk/

design — every component must be self-contained and fit within its dimensions.

4 Escape Room Implementation

Figure 1 shows pictures of the submarine context and provides insights into our design progress augmenting the submarine with pervasive displays and physical interfaces.

The submarine suitcase plays a central role as both a physical anchor for the experience and the game narrative. Historically, this type of suitcase was the only personal belonging that submarine crew members were allowed to bring aboard during each deployment. Everything they needed, documents, personal belongings, a pair or two of underwear and other essentials, had to all fit within this single suitcase. This strict limitation shaped life on board and inspired the core design concept for this escape room experience. A compact suitcase, a central tool of the experience, contains everything participants need to navigate the submarine environment. A benefit of focusing on the suitcase was reducing challenges of designing experiences for and across multiple devices [33]. This section details the development of the escape room, beginning with the distribution of puzzles and the intended game flow. It then outlines the construction of the suitcase and its internal compartments and the five integrated puzzles. In order to encourage others to take up this work in the future, we make the code available in a public repository. [2]

4.1 Puzzle Distribution and Game Flow

This subsection outlines how the puzzles are distributed between the submarine interior and the suitcase, and describes the intended flow of the escape room experience as seen in Figure 2.

4.1.1 Internal & External puzzles. To maintain a balance between gameplay and practical logistics, the escape room was designed with a mix of physical puzzles placed in the submarine's interior as well as the suitcase containing custom-built puzzle elements. This distribution was shaped by the museum's request for a setup that allows quick reset times between groups and minimal intrusion on the Experience Center environment. The puzzles are divided into two main categories:

- **Internal puzzles**: These are embedded in a specially designed suitcase that the players interact with throughout the game. These puzzles are primarily electronic and physically self-contained, but designed to require input from external clues and game elements.
- **External puzzles**: These external clues and game elements are spread across the submarine's three main rooms, requiring players to interact with physical props and items that are part of the environment.

This dual setup aims to strike a balance between gameplay engagement and practical feasibility for the Experience center staff as mentioned in section 3. By incorporating elements of the authentic setting, these puzzles aim to enhance the narrative and encourage exploration.

[2]https://github.com/Johangad/Escape-room-Arduino-box-puzzles-code.git

4.1.2 Game Flow. The escape room is divided into three physical spaces within the submarine, the **Torpedo Room**, the **Command Room**, and the **Engine Room**, as illustrated in Figure 2. The purple section represents the initial suitcase elements available from the start. The green, yellow, and red areas correspond to submarine rooms that become accessible as players unlock key points (e.g., locks D and E) to progress through the space and overall mission. The game is designed to follow a progression like a networked user flow, that allows participants to navigate dynamically between the different spaces. The suitcase, which serves as the starting point and central hub for all internal puzzles, is fixed in the Torpedo Room during the whole experience. Rather than progressing through puzzles in a fixed sequence, players encounter a web of interconnected challenges where artifacts, keys, and clues discovered in one area often unlock or support puzzles in another. This structure aims to promote exploration and collaboration, with players revisiting locations and combining clues across rooms. This also supports parallel puzzle-solving while maintaining a path toward the final goal.

The game begins in the Torpedo Room, which serves as the entry point and narrative starting place. Set in the year **1987**, the story unfolds during the final years of the Cold War, immersing players in a mission aboard a Danish submarine on high alert. Players are introduced to the suitcase and receive a briefing document initializing the experience, a pen and paper, along with a bunch of keys. Here, players encounter the first set of puzzles, which are all the internal puzzles within the suitcase, exposing players to the design and functionalities of the suitcase.

At this stage, players encounter the parallel puzzle-solving format, where they theoretically have access to all puzzles from the beginning. However, they must interact with their surroundings and external puzzle elements to make progress. They have the freedom to begin limited preliminary tasks, which can only be fully solved or utilized later as more information becomes available. The bunch of keys serves as the first guiding element on their escape room journey, initiating their path forward. They will use *key C* and unlock a compartment in their current environment, limited to the Torpedo Room at this point in time. As seen in Figure 2 they will receive elements serving as hints which again will guide them on their path forward, this time by interacting with the internal puzzles for the first time.

This structure is replicated throughout the remaining rooms of the submarine, where players progressively unlock external elements that are essential for solving the internal suitcase puzzles, ultimately guiding them toward mission completion and escape. The structure is further elaborated later under each individual puzzle description.

4.2 Suitcase Construction

A custom insert module was built to fit inside the suitcase, organizing it into distinct compartments while housing the necessary technical components. The module was designed in accordance with the design rationale outlined in section 3, specifically Rationale 3.2, which emphasizes leveraging existing assets and identity by incorporating a real historical artifact into the experience. To align with this, the insert was carefully designed to preserve the

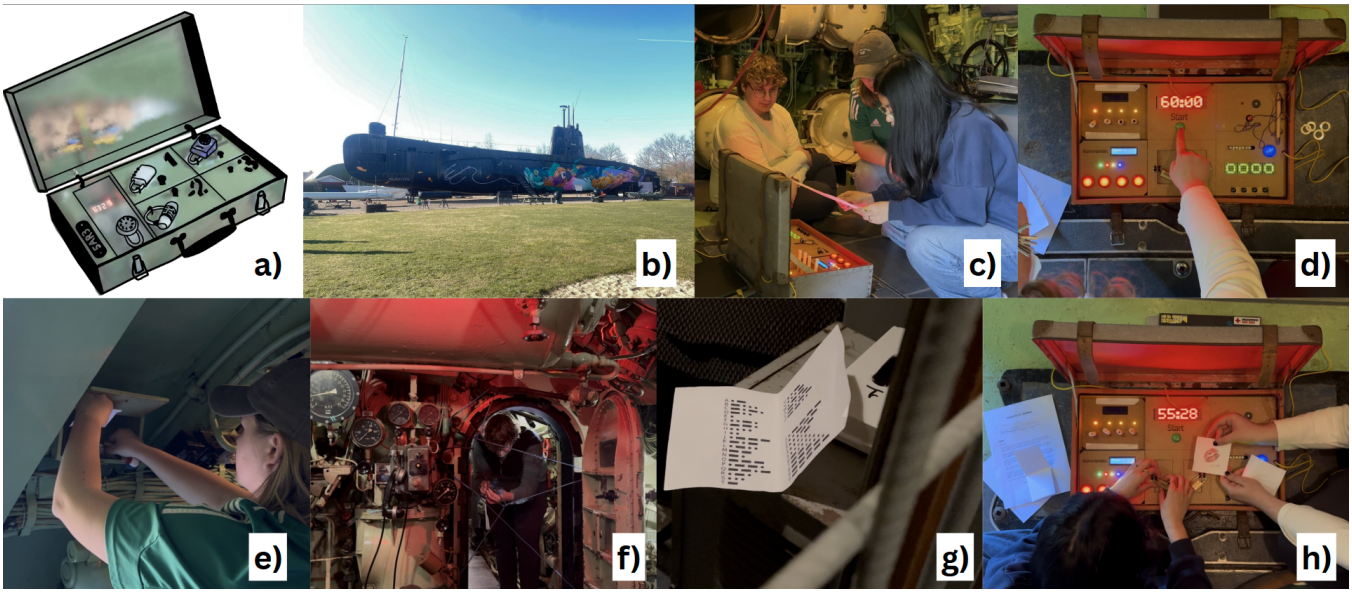


Figure 1: Overview of escape room experience. a) Initial concept sketch of the interactive suitcase. b) Springeren, the real-life setting for the escape room experience. c) Players are introduced to the suitcase and collectively read the introductory letter. d) Escape room experience is initiated by the players pressing the green start button on the suitcase. e) Player explores the environment and unlocks compartment. f) Player gains access to a new area. g) Artifact hidden within the escape room environment. h) Players use artifact to unlock a new compartment in the suitcase.

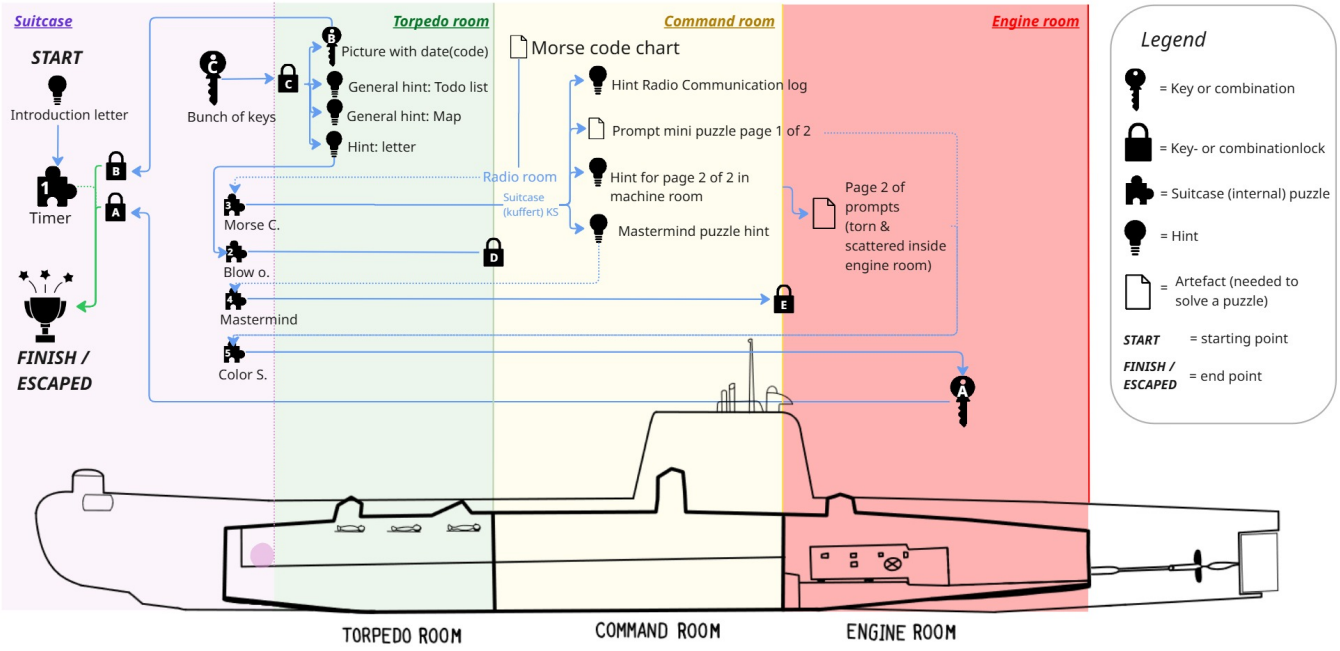


Figure 2: Escape Room Player Flow

suitcase's structural integrity and historical value, ensuring that no damage would be done to the original artifact during construction or gameplay.

The suitcase is divided into six main compartments, each corresponding to a specific puzzle. In alignment with traditional escape room conventions, five puzzles were developed. This structure was chosen based on the assessment that it would fit within the typical 60-minute duration of an escape room experiences. The five puzzles included in the suitcase are: Countdown Timer, Blow-out-the-cigarette, Morse code, Mastermind, and Color sequence.

The insert module of the suitcase was built using MDF, chosen for its sturdiness, low cost, and compatibility with laser cutting. Because the bottom of the suitcase is uneven, the insert had to be designed as a self-supporting box structure rather than a simple frame. After consulting a local makerspace, the module was designed using the TypeTray template from Boxes.py [1], ensuring an accurate and secure structure that was easy to assemble, due to the 'finger joints', which consist of small indents and tabs. A 3×2 equally sized compartment layout was chosen to match the suitcase dimensions and accommodate the six puzzle compartments, including two reserved for the countdown timer.



Figure 3: Image of the fully assembled suitcase, with compartments numbered according to the Escape Room Player Flow in Figure 2.

5 Evaluate and Design-In-Play Cases

Our evaluation and design process is human-centered and tightly coupled, and thus, best described as evaluate-and-design-in-play cases, an adoption of the design-in-use lens [20, 24]. Play tests were conducted and followed by semi-structured interviews. Each case was treated as an individual case study, allowing for design changes between each case based on the previous feedback. This resulted in evolving insights across play tests. The primary focus for the cases was to identify, evaluate and improve the general flow of the escape room, and the internal puzzles. The tests provided insight into the functionality, and emotional aspect of the engagement of the play testers.

5.1 Experimental Procedure

The escape room was tested by 3 different groups consisting of 4 play testers each. These groups had varying prior experience of escape rooms, and how well they knew each other beforehand.

During the play test, data collection tools consisted of note-taking, while using the 'Fly on the wall' approach in the submarine. In the post-test semi-structured interview, notes and audio recordings were used and the interview was primarily based on the PXI model [8]. This was done to further understand the quality of the player experience.

Each play test session began with a brief introduction, welcome outside the submarine, and informed regarding the purpose of the test. The structure of the playtest was open-ended, allowing the play testers to freely engage, and explore within the escape room, whilst a "fly on the wall" observer was present in each room as it was opened.

After the play testers had completed the escape room, they participated in a semi-structured interview. This was to address their overall experience with the escape room, their understanding of the escape room puzzles, and feeling of the general flow. In these overall themes the goal was to find out what aspects worked well and poorly and their perceptions of immersion and learning, throughout the escape room. The participants were also encouraged to offer any suggestions for improvement.

Data consisted of (i) structured observation notes from "fly-on-the-wall" researchers positioned in the submarine spaces as they became accessible, (ii) audio recordings and notes from post-game semi-structured interviews, and (iii) artifact traces (e.g., written notes produced by participants during play and timestamps for major progression events). Interviews were conducted as a group and lasted typically 15–20 minutes.

We analyzed the data using a lightweight form of reflexive thematic analysis [9], suited for the rapid, iterative design case work of the game evaluations. Two researchers reviewed the interview notes and field notes/observations after each session and synthesized the salient patterns directly into a shared document, focusing on recurring breakdowns, successful interaction patterns, and evidence of the target experience (e.g., immersion, collaboration, sense of progression). We also took a deductive approach in our analysis where we considered the design rationales that initiated our design. We gathered inductive insights by capturing patterns noticed across sessions (e.g., role-taking as "crew," leadership/passivity dynamics, perceived urgency). We revisited themes across the three cases and translated them into actionable design updates between sessions, reflecting iterative design-in-use / evaluate-and-redesign case traditions in HCI [20, 24].

5.2 Case A: First Test in the Submarine

Case A, consisted of 4 participants between the ages of 24 to 28. This group was rather experienced, with 3 of the participants having tried 3 or more physical escape rooms and one having tried 1 physical and 1 virtual. The relationship between the participants was also rather close. It is important to note that the puzzle 'Blow-out-the-cigarette' was not included in this round of testing, because it was not working reliably.

Table 1: Operationalization of design rationales in the evaluation.

Design rationale	How it was evaluated in play tests
Ease of installation/removal	Observed reset actions between groups (packing/unpacking, restoring props, re-locking), and noted which elements would require staff attention.
Leveraging existing assets/identity	Interview prompts about perceived authenticity and whether the submarine environment contributed to engagement; observation of exploration behaviors tied to submarine features (e.g. ducking, moving through hatches).
Immersive play as learning	Interview prompts about what participants felt they learned/understood about submarine life; observation of how story cues guided attention to artifacts.
Bridging the age gap	Reported enjoyment, willingness to recommend to peers, and whether the format felt aligned with young adult leisure expectations.
Escape room conventions	Completion within 60 minutes, perceived difficulty, and clarity of start/end states; breakdowns where conventions were violated (e.g., opening the lid).
Submarine as historical artifact	Observed boundary violations (touching restricted areas, opening compartments), and whether instructions/props effectively constrained unsafe or harmful actions.
Design within the suitcase	Observed whether the suitcase functioned as a shared hub (return frequency, coordination around it), and whether puzzles could be solved without modifying the artifact or intervening to give additional information.

The purpose of the first test, is (i) to evaluate whether the technical and spatial setup of the escape room functioned as intended inside the submarine, (ii) to explore how well the flow worked between the technological puzzles in the suitcase and the surrounding analog puzzles, and how the players transitioned between these different elements, and (iii) to determine whether 60 minutes was a realistic duration for the full escape room.

5.2.1 Testing the Setup and Spatial Interaction Inside a Submarine. References to props, and internal and external puzzles throughout the test can be seen in Figure 2. After reading the introduction letter, there was no clear guidance, as to how the escape room would begin the countdown timer. Therefore, there needed to be told to press the green start button. From the beginning, the group demonstrated a confident and exploratory approach. They immediately began testing Key C found in the suitcase, and simultaneously solving the mastermind puzzle before receiving the clue to its use. Thus, making use of the parallel puzzle-solving, mentioned in subsubsection 4.1.2.

In the post-test interview, the players reported that it was mostly clear how things worked. They noted that figuring out what buttons did, was part of the enjoyment they had with the experience. The suitcase worked well as a central hub, they noted, when asked about relocating all the gathered notes throughout the experience at the interactive suitcase. Overall, this indicates that the setup of the escape room, effectively supported independent play and group coordination.

The play testers then found Lock C meant for Key C, they found earlier. Retrieving the hints withheld by the lock, they received a hint to puzzle 2, 'Blow-out-the-cigarette'. Due to the puzzle not working, they received the code to Lock D, a key point in the player flow. Thus, they progressed into Command Room, found the Morse code chart, as seen in Figure 1g), and engaged in a collaborative decoding process and quickly completed puzzle 3, 'Morse code'. The group later in the interview, emphasized that the Morse code puzzle contributed to the immersive experience, in the sense that the play testers got a feeling of being an active crew member. It was also mentioned how having to duck or move differently due to the layout of the submarine further enhanced the feeling of being a crew member.

5.2.2 Flow Between the Internal and External Puzzles. One of the key aspects under the test, was the observation of the transition and balance between the internal puzzles and external puzzles embedded in the environment of the submarine. The flow generally worked, although multiple moments highlighted where refinement could help clarify order or connection.

After completing the Morse code puzzle, the group found and opened 'suitcase KS', relocated its contents to the torpedo room designating it as their central hub. The participants liked that it was easy to have the suitcase as a natural gathering point, so they always knew where to return to. Due to the escape room having parallel puzzle-solving game flow, thus resulting in the play testers always having to go back and forth to progress in the escape room.

In the makeshift central hub the play testers began by collecting the contents of 'Suitcase KS' and quickly identified prompts related to the color sequence puzzle. They attempted to solve it early, despite still needing three crucial command prompts to complete it. Realizing this, they redirected their attention to the engine room and successfully identified the puzzle related to accessing it was the number sequence puzzle. The back and forth progression did not disrupt the experience. Rather, it highlights the escape room's parallel puzzle-solving flow, allowing play testers to explore and engage with multiple challenges simultaneously or out of order. However, it also illustrated that the flow between discovery and execution should be closely monitored in future observations. There must be a careful balance between allowing exploration and participants still needing to follow the escape room player flow to escape

Later, the group efficiently gathered the hints with the last prompts after entering the engine room, even writing them onto one player's arm. After solving the color sequence puzzle, they commented that the display was difficult to read. Additionally, one participant noted that the torpedo room felt 'too open', potentially reducing the atmospheric tension that had been strong earlier in the experience, thus not reflecting that it is nearing the end.

The play test group completed the full experience, excluding one puzzle, with approximately 13 minutes remaining, suggesting that the one-hour time limit seemed realistic.

5.2.3 Identified Issues and Reflection. Some issues identified throughout the experience were

- The play testers' ability to engage with puzzles in a flexible order supported exploration, which is a natural aspect of escape room gameplay. A balance must be maintained between open-ended exploration and enough narrative or structural guidance to prevent unnecessary confusion.
- The play testers were confused about how the countdown timer would begin, and thus start the whole escape room. Therefore, there is a need for clear instructions, so the play testers know to press the green start button to begin the escape room.
- Players hesitated and struggled with finding Key A to unlock the second room for the red button in the suitcase. Some changes to the physical task setup, such as changing the location of the key, may be needed in order to reduce uncertainty and hesitation.
- The LCD mentioned in the Color sequence puzzle was difficult to read. This was linked to the power supply issue, highlighting the importance of testing hardware stability beforehand.
- The group was unsure when the experience ended.

5.3 Case B: All Puzzles Working and Mixed Experience Group

Case B consisted of 4 play testers aged 21-26. This group had mixed levels of prior experience with escape rooms, one having tried one physical, one having tried 5 physical escape rooms, one not having tried any, one having played one, but also played board games and VR equivalents. Compared to case A, their internal group dynamics seemed less cohesive, and this reflected in their communication and task sharing throughout the session.

Some changes were made including the completion of the 'Blow-out-the-cigarette' puzzle, along with changes to props which would allow for observations of improvements and performance under full conditions.

- **Addition of 'Blow-out-the-cigarette':** This session marked the first complete narrative, where all internal puzzles were able to be tested in succession of each other. Thus, allowing to test the whole escape room player flow, and if the 60 minutes will be exceeded or met.
- **Changed feedback for color sequence puzzle:** Because the play testers from Case A had a hard time locating the Key A, a small change was added to the hint given by solving the color sequence puzzle, the text which previously read: "Behind the gauge, big red wheel", to "Behind the gauge, *BY* the big red wheel". This small change is based on the hint the play testers were given, leading them to find Key A.
- **Display had power issues:** The dimming of the LCD screen reported in the previous case was fixed by buying completely new 9V batteries, assuming that replacing the power supply had resolved the issue.
- **Changes to props** Changes to some props were needed to address the issues identified in Case A. These changes include more structural guidance to begin the escape room. Therefore, the intro document given to the play testers was altered, in which a message at the bottom was added that reads: "Press 'start' to initiate the escape room". This guidance should make sure the play testers knew exactly when the game began. Furthermore, the location of the Key A was made easier to spot, based upon the observation of the play test group in Case A. Since they quickly found the right location of the key but could not find the key itself, as it was too well hidden. It was decided to make the key more visible by allowing it to peek out.

This second case aimed to test the full experience of the escape room, with all puzzles being active. This included the 'Blow-out-the-cigarette' puzzle, which was not implemented during Case A. This session offered a great opportunity to observe the complete setup performed in practice, and how a group with mixed experience, compared to the play testers in Case A, experienced the difficulty of the experience.

The core objectives were to evaluate (i) how well the escape room functioned when all puzzles were in place, (ii) to observe how a mixed-experience group collaborated and moved between the puzzles, and (iii) assess whether the 60-minute time limit remained appropriate with all the puzzles in place.

The group quickly divided tasks, with the most experienced player taking a persistent, non-rotating lead by operating and interpreting the suitcase. This centralized control reduced shared understanding: key information (e.g., the first letter and parts of "Suitcase KS") was not read aloud or fully communicated, and puzzles were sometimes solved in isolation, leaving one participant passive in the final 20 minutes. Although curiosity and movement between puzzles remained high, missteps such as repeatedly reusing the same key bundle created confusion. The session ran about 7 minutes over the intended 60 minutes, partly due to a mid-session pause to repair suitcase wiring.

5.4 Case C: Integrated Evaluation and Robustness

Case C consisted of four play testers aged 22-24. This group was somewhat experienced with escape rooms, wherein all of the play testers had tried escape rooms before, but the amount of experience was varied. Two of the play testers both tried three escape rooms, whereas the other two play testers had only attended one escape room before. The relationship between the participants was also rather close, as three of the participants all attend the same education and semester, and the fourth member attended the same education but on a different semester.

Based on the two other cases, the focus was finding participants that knew each other beforehand, this was done in order to see if there could be a correlation between how well the play testers did, based on them knowing each other, and if factors like that had any role in the play testers' ability to work together. Additionally focusing on making the suitcase more power supply stable as well as robust, which were problems encountered from Case B. The reasoning behind making the prototype more stable, was to ensure

that all of the puzzles and technology were working as intended, in order to get a full picture with more accurate timestamps for the entire escape room.

The importance of collaboration and teamwork was also more apparent during the observations of Case C, as the group followed the instruction, which put emphasis on the importance of working together and reading clues aloud. This was a similar dynamic observed with players in Case A, which was also a very collaborative group.

The play testers responded that they found the escape room fun, but it was hard to get going in the beginning.

The play test group in Case C, like the play test group in Case B, were also looking for some more immersion by adding audio in the submarine. The play test group also responded that they would like some more "pressure" added to the escape room experience, since they did not get the feeling of urgency throughout the escape room.

5.5 Findings Across Cases

Collaboration and Group Dynamics. Case A demonstrated highly effective collaboration, likely influenced by the pre-existing relationship within the group and their experience with prior escape rooms. As mentioned in subsection 5.2, the participants communicated efficiently, divided tasks and showed confidence in navigating the space inside the submarine. In contrast, **Case B** exhibited weaker internal dynamics, as mentioned in subsection 5.3. Although the players initially attempted to collaborate, confusion and overlapping actions happened. This was partly due to the unclear communication within and rule violations, especially regarding interaction with the suitcase. **Case C** while more aligned with Case A in terms of coordination, showed a more hesitant approach, possibly due to limited prior experience. However the players in Case C adapted quickly, and their collaboration improved as confidence grew, throughout the experience.

The play testers in Case C, ended up solving the escape room in about the same amount of time as the play testers in Case A, even though the play testers in Case C had to solve the 'Blow-out-the-cigarette' and were stuck on the puzzle for around 20 minutes. The play testers in Case A, were however also stuck on the last part of finding the Key A for the red button, which accounted for around 15 minutes, which was made easier throughout Case B and C. The play testers in Case A and Case C, both solved the puzzles in around 40-45 minutes. Whereas the play testers in Group B did not solve the puzzle in the allotted time slot of 60 minutes. This seems to somewhat provide a basis on which future work could be done, to more confidently conclude if the play testers knowing each other, plays a role in how well a group can solve the escape room before the time runs out.

Player Flow and Progression. Each case highlighted different challenges in the progression of the puzzles. In case A, some tasks, such as finding the Key A, proved too difficult, along with the 'Blow-out-the-cigarette' puzzle breaking, thus allowing for the puzzle to be bypassed, and breaking the intended flow of the escape room. Case B included all the puzzles, which was intended to provide a full test of the intended flow, however due to accidental tampering of the suitcase by the players, caused technical malfunctions. Case C, benefited from these findings, clearer instructions, more robust prototype, update of props and hardware adjustments, helped maintain the designed flow more effectively, and all the puzzles remained functional throughout, however Case C also showed some lack of fine tuning in the difficulty of specific puzzles, such as the 'Blow-out-the-cigarettes', which the play test group struggled to solve.

Time Management and Pacing. The time constraints were differently experienced across the groups. Case A and C both completed the experience with approximately 15 minutes left, while B exceeded the time by 7 minutes. These variances could suggest the pacing of the experience can change depending on the group composition and stability of the puzzles.

Rule Comprehension A recurring theme across the cases was how well the participants understood the rules and respected the boundaries for the setup, particularly regarding the suitcase. Case A and C followed these instructions closely, while B repeatedly violated the instruction to not open the lid into the suitcase, causing puzzle failure. Thus, highlighting the importance of balance between freedom of exploration and structural guidance.

6 Discussion

Visionaries in HCI have long speculated how pervasive computing and tangible interaction can reshape how people experience their surrounding world [15, 18]. Our design case extends this agenda in a museum heritage setting by treating the submarine not as a backdrop, but as an interactionally consequential place whose material constraints and cultural meanings shape play.

6.1 Interpreting the Experience through Embodied Place-based Interaction

Using an embodied interaction lens [15], the submarine's constrained interior becomes an active ingredient of the experience rather than a neutral backdrop. This interpretation is grounded in observations from Case A and Case C where participants' reports of "feeling like crew" emerged when bodily navigation (ducking, squeezing past fixtures) aligned with narrative prompts and collaborative task coordination. This suggests that immersion here is partly produced by *place-based enactment*: the physical demands of movement and proximity structure attention, communication, and role-taking in ways difficult to reproduce in a generic room.

While the confined space supported embodied role-taking, participants also pointed to opportunities for strengthening atmosphere through additional sensory cues. In particular, in post-test interviews, participants suggested that ambient audio (e.g., sonar pings, engine hum, emergency announcements) could heighten urgency and narrative coherence, complementing the physical setting. Additionally, our "fly-on-the-wall" presence inside the submarine may have reduced perceived theatrical tension; future evaluations could use remote monitoring (e.g., fixed cameras) to reduce observer intrusiveness while enabling richer interaction analysis.

6.2 Implications for Designing Escape Rooms in Heritage Environments

Our cases highlight tensions specific to heritage settings. First, the *freedom to explore* that makes escape rooms enjoyable can conflict with preservation and safety constraints. Second, a distributed environment can support discovery, but risks fragmentation when groups split, information is not shared, or conventions are violated (e.g., opening protected compartments). This tension was visible in Case B, where one participant monopolized the suitcase interface, solving puzzles in isolation, which reduced shared understanding and left another participant passive. Third, operational feasibility (reset time, robustness of electronics, clarity of start/end states) is not peripheral; it shapes whether such experiences can scale beyond a one-off installation.

A parallel direction in museum innovation explores digital and sensory augmentation to increase relevance and participation for young adult visitors. For example, the Kunsten project investigates digital and sensory-stimulating installations in an art-museum context as a strategy for strengthening young adults' engagement [4]. We position our work as complementary: rather than primarily augmenting exhibits through digital overlays, we leverage an embodied, place-based heritage environment and time-bounded collaborative play to structure exploration and meaning-making in situ.

6.3 Design Guidelines

Based on cross-case findings and design rationales, we propose the following guidelines for escape-room experiences in museums:

(1) **Exploit embodied constraints as interaction resources.** In tightly constrained heritage spaces, design for movement, posture, and proximity to support role-taking and immersion (e.g., tasks that require shared viewing or coordinated passing of artifacts), rather than treating constraints as obstacles.
(2) **Use a physical hub to orchestrate distributed interaction.** A central artifact (here, the suitcase) can stabilize collaboration by providing a shared point for regrouping, combining clues, and confirming progress. Ensure that hub interactions are legible to bystanders (displays readable from multiple angles; feedback not limited to a single operator).
(3) **Balance parallelism with explicit re-synchronization points.** Parallel puzzle-solving supports agency and exploration, but add periodic "everyone together" moments (e.g., unlock events, briefing updates) that re-align shared understanding and reduce silent solo-solving.
(4) **Make start/end states unambiguous.** Confusion about how to start and when the experience ended appeared in early cases. Use redundant cues (printed instruction + physical affordance + audiovisual confirmation) to mark initiation and completion.
(5) **Design boundaries as part of the narrative.** Preservation/safety constraints should be communicated diegetically (e.g., "sealed compartments" framed as protocol) so that restrictions feel like story-consistent rules rather than external prohibitions.
(6) **Engineer for robustness and fast reset from the outset.** In museum operations, the experience must tolerate misuse and variability. Favor components that fail gracefully, make wiring inaccessible, and provide a reset checklist that staff can execute quickly.

6.4 Limitations and Transferability

This paper reports an in-situ design case with $n = 12$ participants across three groups. While the submarine context is distinctive, the guidelines above are transferable to other heritage environments that combine (i) constrained authentic spaces, (ii) strict preservation constraints, and (iii) small-group, time-bounded playful interaction. Future work should evaluate through additional sessions at the museum and examine visitor experiences across a wider visitor profile and puzzle permutations.

7 Conclusion

The development and implementation of the submarine escape room demonstrate a possible approach to revitalizing cultural spaces for younger audiences. By transforming the museum's submarine into a space for interactive, social, and immersive play, the project contributes to how museums might deliver historical narratives in ways that resonate with young adults' expectations for museums. The design emphasizes modularity, authenticity, and narrative coherence, ensuring that the experience remains respectful of the submarine's historical heritage while inviting playful engagement.

The use of pervasive technologies, tangible interactions, conversational user interfaces, and authentic physical artifacts, notably the submarine suitcase, creates an environment where learning and entertainment intersect. Feedback from iterative playtesting revealed a high level of user engagement, and satisfaction with the puzzle design and narrative integration. Challenges such as unclear starting instructions and ending cue feedback were identified and addressed between test sessions, further refining the user experience. We hope this work inspires others to envision museums as playful, participatory spaces that invite young audiences to engage meaningfully with historical artifacts.

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